

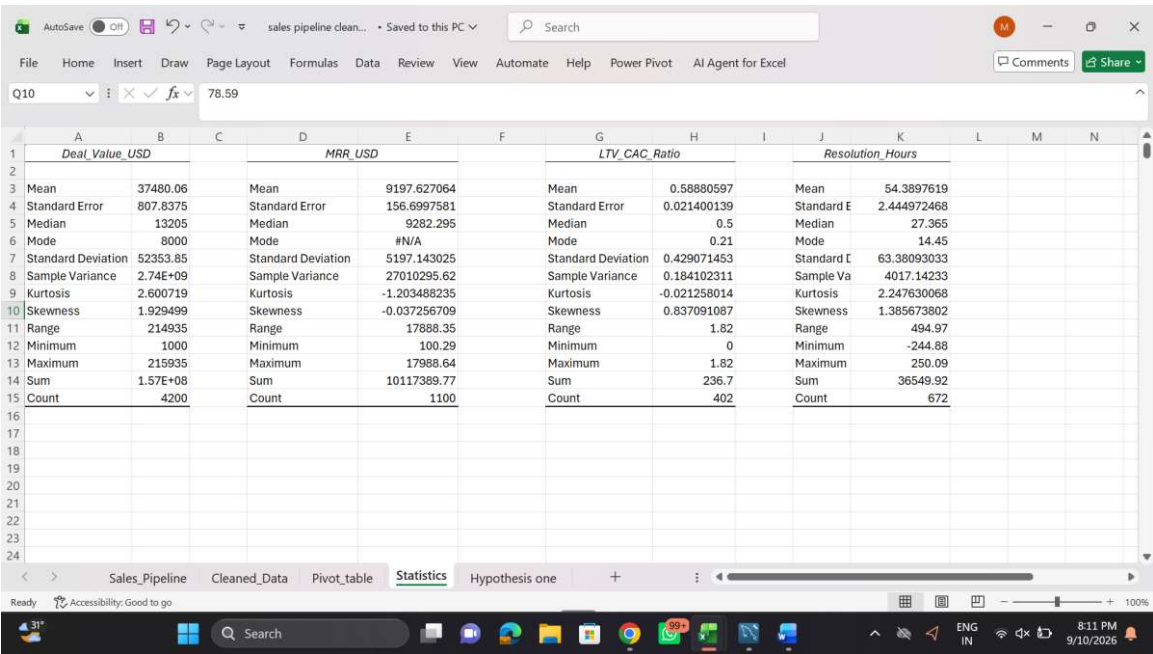
Task 5: Statistical Validation Report

Project: Gen P Technologies - Business Analytics Capstone

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This report documents the baseline descriptive statistics of key business metrics and the statistical stress-testing of initial business observations. Through rigorous hypothesis testing, we validate which trends represent true operational dynamics versus random variance.

1. Baseline Descriptive Statistics (Business Interpretation)



The screenshot shows an Excel spreadsheet with four columns of statistical data. The columns are labeled: Deal Value_USD, MRR_USD, LTV_CAC_Ratio, and Resolution_Hours. Each column contains a list of statistical measures and their corresponding values. The measures include Mean, Standard Error, Median, Mode, Standard Deviation, Sample Variance, Kurtosis, Skewness, Range, Minimum, Maximum, Sum, and Count. The values are displayed in a clean, professional format with appropriate decimal places and scientific notation where necessary.

	Deal Value_USD	MRR_USD	LTV_CAC_Ratio	Resolution_Hours
Mean	37480.06	9197.627064	0.58880597	54.3897619
Standard Error	807.8375	156.6997581	0.021400139	2.444972468
Median	13205	9282.295	0.5	27.365
Mode	8000	#N/A	0.21	14.45
Standard Deviation	52353.85	5197.143025	0.429071453	63.38093033
Sample Variance	2.74E+09	27010295.62	0.184102311	4017.14233
Kurtosis	2.600719	-1.203488235	-0.021258014	2.247630068
Skewness	1.929499	-0.037256709	0.837091087	1.385673802
Range	214935	17888.35	1.82	494.97
Minimum	1000	100.29	0	-244.88
Maximum	215935	17988.64	1.82	250.09
Sum	1.57E+08	10117389.77	236.7	36549.92
Count	4200	1100	402	672

Deal Value (USD)

- **Key Metrics:** Mean: \$37,480 | Median: \$13,205 | Skewness: 1.93

Business Insight: The deal value data is highly right-skewed. While the "typical" contract is smaller (around \$13,205), the company occasionally lands massive "whale" contracts (up to \$215,935) that artificially drag the overall average up to \$37,480. Sales forecasting should rely on the median to avoid overestimating standard pipeline revenue.

Monthly Recurring Revenue (MRR_USD)

- **Key Metrics:** Mean: \$9,197 | Median: \$9,282 | Skewness: -0.04 | Kurtosis: -1.20

Business Insight: MRR is perfectly symmetrical and highly predictable. The average and median are nearly identical. The negative kurtosis indicates a "flat" distribution, meaning recurring revenue is spread out evenly across various pricing tiers rather than tightly clustered around a single standard package.

Unit Economics (LTV_CAC_Ratio)

- **Key Metrics:** Mean: 0.59 | Median: 0.50

Business Insight: This mathematically confirms a severe profitability squeeze. A healthy SaaS company targets a ratio of 3.0 or higher. With an average ratio of 0.59, the company is systematically spending nearly twice as much to acquire customers as they will ever make back in lifetime value.

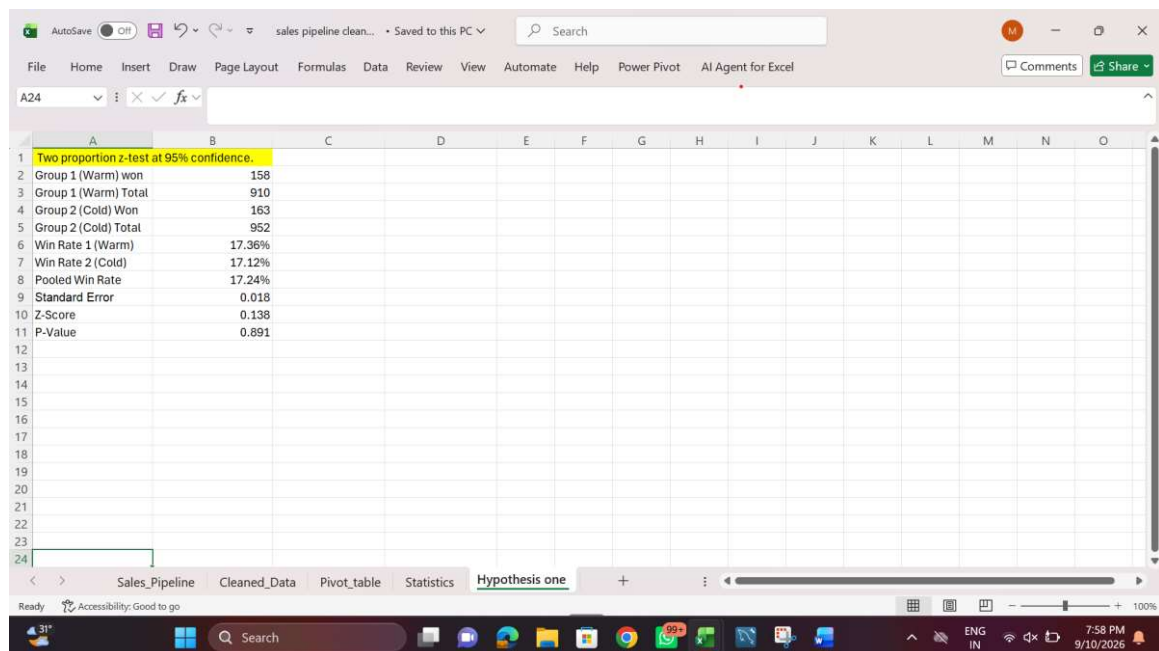
Support Ticket Resolution Hours

- **Key Metrics:** Mean: 54.39 | Median: 27.37 | Skewness: 1.39

Business Insight: Similar to Deal Value, support times are heavily right-skewed. The typical ticket is resolved in about 27 hours. However, the average is dragged up to 54 hours by a small group of extremely difficult tickets that remain open for days (maximum of 250 hours).

2. Hypothesis Testing Results

Hypothesis 1: Deal Source Win Rates (Two-Proportion Z-Test)

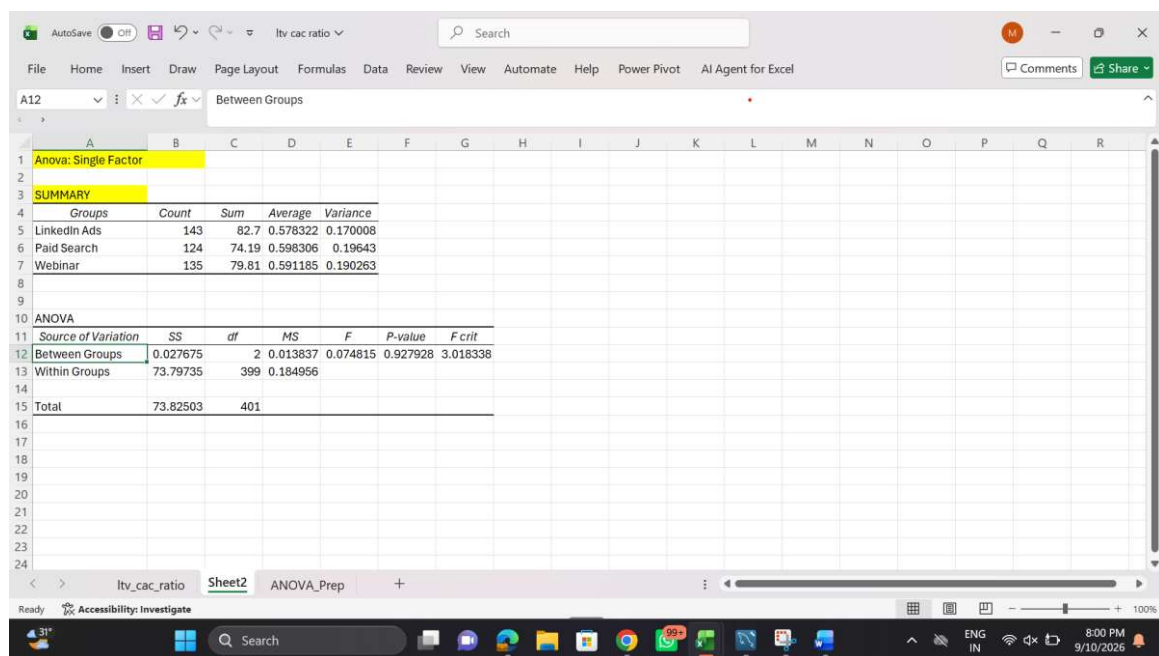


1	Two proportion z-test at 95% confidence.	
2	Group 1 (Warm) won	158
3	Group 1 (Warm) Total	910
4	Group 2 (Cold) Won	163
5	Group 2 (Cold) Total	952
6	Win Rate 1 (Warm)	17.36%
7	Win Rate 2 (Cold)	17.12%
8	Pooled Win Rate	17.24%
9	Standard Error	0.018
10	Z-Score	0.138
11	P-Value	0.891

- **Null Hypothesis:** Win rates for Product-Led/Organic leads and Cold/Paid leads are equal.
- **Test Statistic (Z-Score):** 0.137
- **P-Value:** 0.89
- **Decision:** Fail to reject the null hypothesis.

Business Interpretation: Win rates for "warm" leads (17.36%) and "cold" leads (17.12%) are statistically identical. The sales team converts cold outreach just as effectively as inbound organic traffic.

Hypothesis 2: LTV/CAC Ratio by Channel (One-Way ANOVA)



The screenshot shows an Excel spreadsheet titled "ltv_cac_ratio" with a worksheet named "ANOVA_Prep". The spreadsheet contains two tables: a summary table and an ANOVA table.

Summary Table:

Groups	Count	Sum	Average	Variance
LinkedIn Ads	143	82.7	0.578322	0.170008
Paid Search	124	74.19	0.598306	0.19643
Webinar	135	79.81	0.591185	0.190263

ANOVA Table:

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	0.027675	2	0.013837	0.074815	0.927928	3.018338
Within Groups	73.79735	399	0.184956			
Total	73.82503	401				

- **Null Hypothesis:** Average LTV/CAC ratios are equal across all acquisition channels.
- **Test Statistic (F):** 0.074
- **P-Value:** 0.927
- **Decision:** Fail to reject the null hypothesis.

Business Interpretation: The profitability squeeze is universal. No single channel (Webinar, LinkedIn, Paid Search) is significantly outperforming the others regarding customer acquisition economics; all hover around a poor 0.58 average ratio.

Hypothesis 3: SLA Breaches and Churn Risk (Chi-Square Test)

The screenshot shows an Excel spreadsheet titled 'Support_Ticket...' with a 'Chi-square test' tab selected. The spreadsheet contains two tables: 'Observed Grid' and 'Expected Grid'.

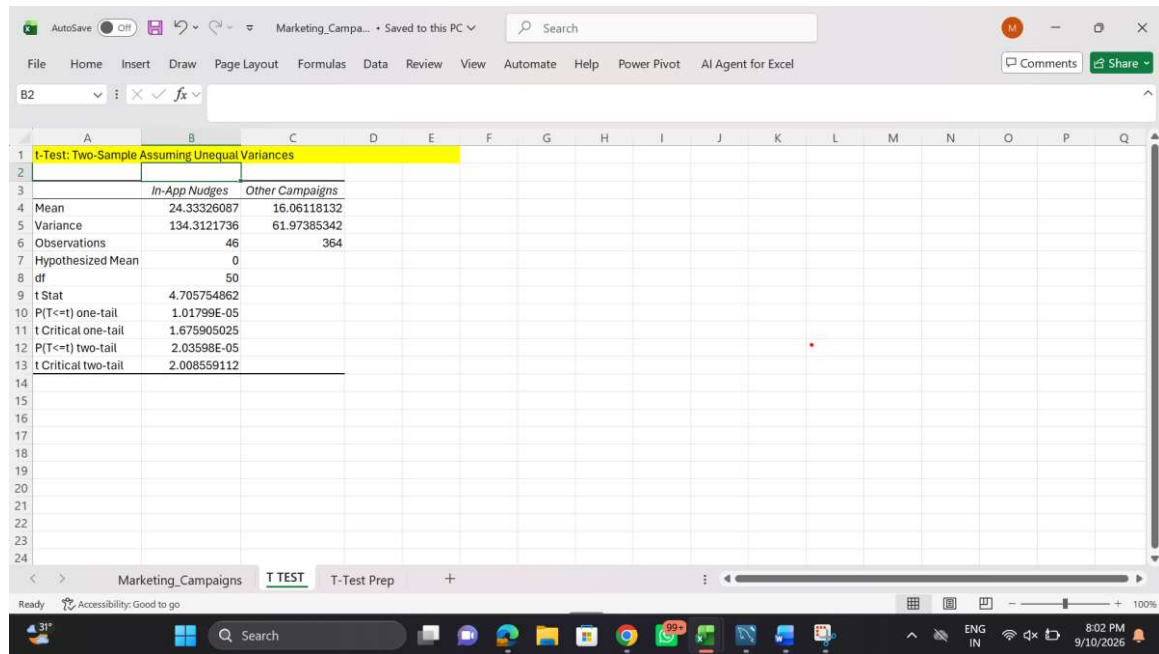
	High Churn	Low Churn	Total
Breached SLA	447	888	1335
Met SLA	464	937	1401
Total	911	1825	2736

	High Churn	Low Churn
Breached SLA	444.51206	890.48794
Met SLA	466.48794	934.51206
P-Value	0.839985	

- **Null Hypothesis:** Churn risk is independent of whether a customer experienced an SLA breach.
- **P-Value:** 0.84
- **Decision:** Fail to reject the null hypothesis.

Business Interpretation: Experiencing an SLA breach does not statistically increase a customer's likelihood of churning. Customer service response times are not the primary driver of customer attrition.

Hypothesis 4: Campaign Activation Rates (Independent Samples T-Test)



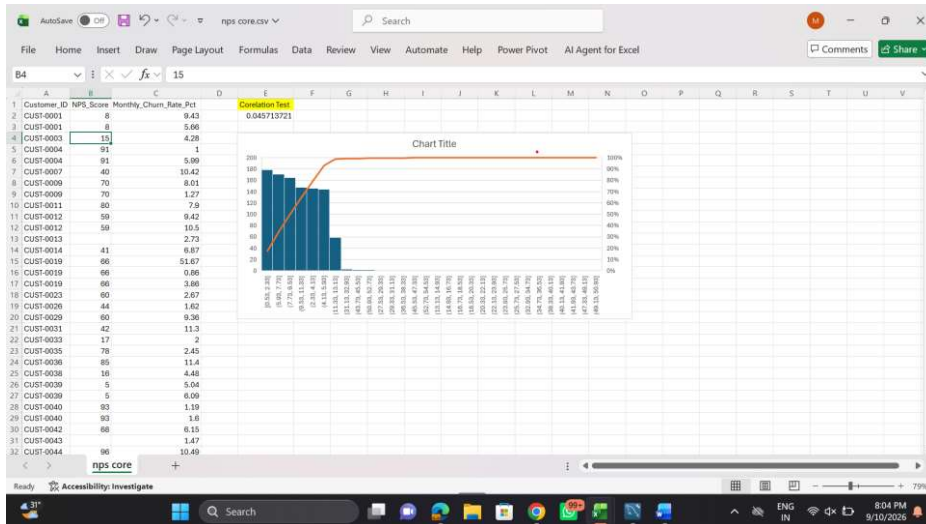
t-Test: Two-Sample Assuming Unequal Variances		
	In-App Nudges	Other Campaigns
Mean	24.33326087	16.06118132
Variance	134.3121736	61.97385342
Observations	46	364
Hypothesized Mean	0	
df	50	
t Stat	4.705754862	
P(T<=t) one-tail	1.01799E-05	
t Critical one-tail	1.675905025	
P(T<=t) two-tail	2.03598E-05	
t Critical two-tail	2.008559112	

- **Null Hypothesis:** In-App Nudge campaigns and other campaigns have equal activation rates.
- **Test Statistic (t):** 4.71
- **P-Value:** 0.00002
- **Decision:** Reject the null hypothesis.

Business Interpretation: In-App Nudges drive a statistically significant increase in activation rates (24.33%) compared to other campaigns (16.06%). Marketing resources should be heavily reallocated toward in-app onboarding flows.

2. Correlation and Distribution Diagnostics

NPS Score vs. Monthly Churn Rate (Pearson Correlation)



- **Correlation Coefficient (r):** 0.045

Business Interpretation: There is zero statistical relationship between a customer's Net Promoter Score and their Monthly Churn Rate. A high NPS does not protect the company from churn, indicating that while customers may like the product, it lacks the "stickiness" required for retention.

Distribution of Monthly Churn Rate

- **Shape Identified:** Right-Skewed (Positively Skewed).

Business Interpretation: The histogram analysis reveals a heavy concentration of customers on the lower end of the churn rate scale, with a long tail extending to the right. While the majority of the customer base exhibits stable, low churn, a small but distinct segment of outlier accounts experiences exceptionally high churn rates that require targeted intervention.